

Notes: Benamar, Foucault & Vega (RFS, 2021)

Demand for Information, Uncertainty, and the Response of U.S. Treasury Securities to News

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1 Big picture

- **Question.** Can investors' demand for information be used as a direct, high-frequency proxy for uncertainty, and does it predict how strongly Treasury prices react to macroeconomic news?
- **Core idea.** When investors are more uncertain about the future path of interest rates, they search more for information before announcements that can resolve this uncertainty. This higher information demand should be associated with a larger price response to the announcement surprise.
- **Data innovation.** The paper uses clickstream data from Bitly links to news articles. The main application measures clicks on articles about nonfarm payroll employment ahead of the 8:30 a.m. ET payroll announcement.
- **Asset-pricing setting.** U.S. Treasury notes are ideal because macroeconomic announcements, especially nonfarm payroll, directly affect beliefs about future interest rates and therefore Treasury yields.
- **Main result.** A one-standard-deviation increase in pre-announcement Bitly clicks raises the sensitivity of Treasury futures yields to nonfarm payroll surprises by about 3 bps for 2-year notes, 3.5 bps for 5-year notes, and 2.7 bps for 10-year notes.
- **Robustness.** The relationship survives controls for other uncertainty proxies, information supply, Google search, order flow, trading volume, realized volatility, and zero-lower-bound effects.
- **Identification logic.** The paper does not claim clicks randomly cause larger price responses. Instead, the model predicts that uncertainty shocks jointly raise information demand and price sensitivity to public news.
- **Non-noise interpretation.** High clicks do not simply predict post-announcement reversals or slow adjustment. Price reactions are larger at the announcement and then persist, which supports the uncertainty interpretation rather than retail attention/noise trading.
- **Generalization.** The same logic also appears for other announcements: retail sales, ISM manufacturing PMI, and the path component of FOMC surprises.
- **Takeaway.** Clickstream measures are useful because they reveal what information investors seek right before payoff-relevant news arrives. They can contain uncertainty information not captured by standard market-based or news-based proxies.

2 Data and measurement

2.1 Clickstream data and NFP articles

- The paper obtains clickstream data from Bitly, which records clicks on shortened URLs.
- For the main tests, the authors identify URLs pointing to articles about nonfarm payroll by using keywords such as “payroll” and “jobs report.”
- The resulting set of shortened URLs is called the set of NFP SURLs; clicks on these URLs are NFP clicks.
- The main information-demand variable is the number of NFP clicks in the two hours before the payroll announcement, from 6:29 a.m. to one minute before 8:30 a.m. ET.
- For the core tests, the paper focuses on Bitly clicks to Bloomberg and Wall Street Journal articles to better capture investor demand for financial information.

Interpretation: Clicks are interpreted as revealed demand for information. The timing is crucial: the measure is built before the announcement, so it predicts the reaction to future news rather than mechanically responding to the news.

2.2 Information demand variables

The main variables are:

Bitly count_{*t*} = number of relevant Bitly clicks before announcement *t*,
standardized by its sample standard deviation, and

High Bitly count_{*t*} = $\mathbf{1}\{\text{Bitly count}_t > \text{median Bitly count}\}$.

Interpretation: The continuous variable measures the intensity of information search. The high-count dummy provides a simple above-versus-below-median comparison.

2.3 Treasury yield responses

- The dependent variable is a 30-minute U.S. Treasury futures yield change.
- The window runs from the prevailing futures price one second before the announcement to 30 minutes after the announcement.
- The main maturities are 2-year, 5-year, and 10-year U.S. Treasury note futures.
- Nonfarm payroll surprises are announcement values minus expectations, standardized by the surprise standard deviation.

Interpretation: The short 30-minute window isolates the market’s immediate response to public news and limits contamination from other events.

2.4 Alternative uncertainty and information measures

The paper compares Bitly information demand with:

- a market-based uncertainty measure from interest-rate swaps;
- a news-based uncertainty measure;
- VIX;
- absolute forecast errors;
- dispersion across professional forecasters;
- 2-year Treasury realized volatility;
- Google Trends for nonfarm payroll;
- RavenPack news counts, interpreted as information supply.

Interpretation: Bitly clicks are not just another news-count measure. A key distinction is demand versus supply: clicks measure what investors choose to read, while news counts measure what media outlets produce.

3 Model and empirical specifications

3.1 Public news and private information

The risky asset payoff has a component *F*. At date 2, a public signal arrives:

$$A_e = F + \epsilon. \tag{1}$$

Speculator i can acquire a private signal

$$s_i = F + \eta_i, \quad (2)$$

where η_i has precision $\tau_{\eta i}$. Aggregate information demand is

$$\tau_\eta = \int_i \tau_{\eta i} di. \quad (3)$$

Interpretation: Precision is the model counterpart of information demand: investors search more when the marginal value of information is high.

3.2 Trading and price updating

Aggregate order demand is

$$D(p_1) = \int_i x_i(s_i, p_1) di + u, \quad (4)$$

where u is noise-trader demand. Competitive dealers set

$$p_1 = \mu + E(F \mid D(p_1)), \quad (5)$$

and after public news arrives,

$$p_2 = \mu + E(F \mid D(p_1), A_e). \quad (6)$$

The price response to the public signal can be written as

$$p_2 = p_1 + \beta (A_e - E(A_e \mid z_1)), \quad (7)$$

where the sensitivity of price to the announcement surprise is

$$\beta = \frac{\text{Var}(F \mid z_1)}{\text{Var}(F \mid z_1) + \text{Var}(\epsilon)}. \quad (8)$$

Interpretation: The larger investors' uncertainty about F before the announcement, the more the public signal moves prices. This is the model's direct link between uncertainty and announcement sensitivity.

3.3 Information acquisition

A speculator's certainty equivalent from acquiring signal precision $\tau_{\eta i}$ is

$$CE(\tau_{\eta i}, \tau_\eta) = \frac{1}{2\gamma} [\ln(1 + \tau_{\eta i} \text{Var}(F \mid z_1))] - c(\tau_{\eta i}). \quad (9)$$

Interpretation: Information demand rises when uncertainty increases because information becomes more valuable. But greater aggregate information demand also makes prices more informative, partially offsetting uncertainty.

3.4 Main theoretical implication

The model's main implication is:

uncertainty shock $\Rightarrow$ higher information demand $\Rightarrow$ larger price response to public news.

Interpretation: The model predicts a positive comovement between pre-announcement clicks and announcement price sensitivity. The empirical test is not that clicks cause price movements, but that clicks reveal uncertainty.

3.5 Baseline announcement regression

The baseline response regression is

$$\Delta y_{\tau,t} = \alpha + \beta_S \text{Surprise}_t + \varepsilon_t. \quad (10)$$

Interpretation: β_S is the average response of a Treasury yield of maturity τ to a standardized nonfarm payroll surprise.

3.6 Main interaction regression

The main regression is

$$\Delta y_{\tau,t} = \alpha + \beta_S \text{Surprise}_t + \beta_{SI} \text{Surprise}_t \times I_{dem,t}^{before} + \beta_I I_{dem,t}^{before} + \varepsilon_t. \quad (11)$$

- $I_{dem,t}^{before}$ is Bitly count or High Bitly count before announcement t .
- The key coefficient is β_{SI} .
- The model predicts $\beta_{SI} > 0$.

Interpretation: A positive interaction means the same payroll surprise moves yields more when investors searched more before the announcement.

3.7 Controls and horse-race specification

The paper extends the interaction specification by adding controls and surprise-control interactions:

$$\begin{aligned} \Delta y_{\tau,t} = & \alpha + \beta_S \text{Surprise}_t + \beta_{SI} \text{Surprise}_t \times I_{dem,t}^{before} + \beta_I I_{dem,t}^{before} \\ & + \sum_j \theta_j X_{j,t} + \sum_j \phi_j \text{Surprise}_t \times X_{j,t} + \varepsilon_t. \end{aligned} \quad (12)$$

Interpretation: This tests whether Bitly demand contains uncertainty information beyond standard uncertainty proxies, Google search, information supply, volume, volatility, and monetary-policy controls.

3.8 Pre- and post-announcement reaction

To test for reversals or slow adjustment, the paper estimates daily lead-lag regressions of the form

$$\Delta \text{DailyYield}_t = \alpha + \sum_{i=-30}^{30} \beta_{S,i} \text{Surprise}_{t-i} + \sum_{i=-30}^{30} \beta_{BS,i} \text{Surprise}_{t-i} \times \text{BitlyCount}_{t-i} + \varepsilon_t. \quad (13)$$

Interpretation: If high clicks reflect noise attention, we might expect reversals. If they reflect uncertainty, the price impact should occur at the announcement and persist.

4 Empirical design and credibility

4.1 Why the timing helps

- Information demand is measured before the public announcement.
- The outcome is the price reaction immediately after the announcement.
- The surprise is standardized using expectations, so the test focuses on the market's reaction to unexpected news.

Interpretation: The design separates search-before-news from price reaction-after-news.

4.2 What identifies the mechanism

- The model predicts that uncertainty increases both information demand and sensitivity to public news.
- The empirical interaction between surprise and Bitly count tests this prediction.
- Controls ask whether Bitly is just a proxy for other uncertainty measures or media supply.
- The pre/post tests ask whether high information demand merely captures noise trading or delayed price adjustment.

Interpretation: The paper's strongest evidence is that Bitly demand has incremental explanatory power exactly when and where the theory says uncertainty should matter.

4.3 Limits

- Bitly users are not all investors, and the paper cannot observe every information source investors use.
- The tests are predictive, not a randomized experiment.
- Clicks may reflect a mixture of institutional demand, retail demand, and journalistic diffusion.
- The focus is Treasury-related macro news; other asset classes may behave differently.

Interpretation: The paper is best read as evidence that clickstream data are a useful proxy for uncertainty, not as proof that clicks themselves move Treasury prices.

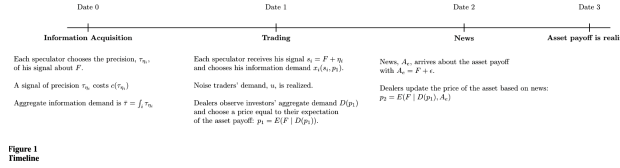
5 Tables and figures

5.1 Figure 1 and Figure 2: model timeline and intraday click pattern

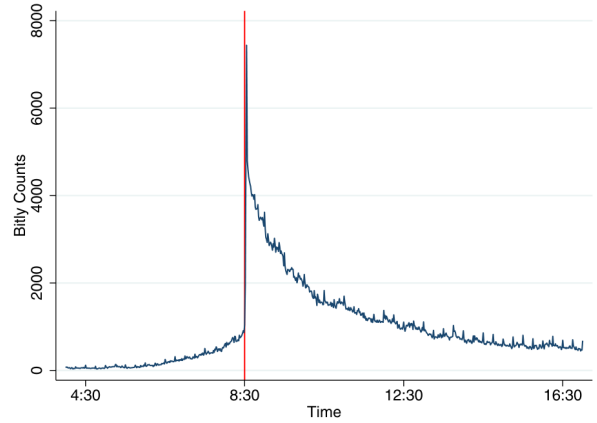
Read:

- Figure 1 summarizes the theory: speculators acquire information, trade, public news arrives, and then the payoff is realized.
- Figure 2 shows that NFP-related Bitly clicks rise gradually before 8:30 a.m. and then spike immediately after the release.
- The pre-announcement rise is important because the paper uses clicks before the release as the uncertainty proxy.

Economic message: The timing supports the paper’s empirical design. Clicks are observed before the information event and can be used to predict how strongly yields respond to the surprise.



(a) Figure 1: model timeline



(b) Figure 2: intraday NFP Bitly clicks

5.2 Tables 1 and 2: who produces and shares the clicked news?

Read:

- Table 1 shows that pre-release clicks are concentrated in financial news sources: Wall Street Journal accounts for 34%, Bloomberg 26%, and CNN 21%.
- After the release, the top five sources still account for 72% of clicks.
- Table 2 shows that official accounts of major news providers create a large fraction of the links that receive clicks.
- These patterns support the interpretation that NFP clicks are linked to financial-news readership rather than random web traffic.

Economic message: The source composition matters because the paper wants to measure investor demand for macro-financial information.

Table 1
Popular news sources of articles shared using Bitly

News source	Number of clicks	Percent of total number of clicks	Cumulative percent
<i>A. Prior to nonfarm payroll release, from 6:29 a.m. ET to 8:29 a.m. ET</i>			
Wall Street Journal	7,137	34	34
Bloomberg	5,538	26	60
CNN	4,365	21	80
New York Times	1,219	6	86
USA Today	1,144	5	91
<i>B. During and after nonfarm payroll release, from 8:30 a.m. ET to 10:30 a.m. ET</i>			
Wall Street Journal	26,200	18	18
CNN	24,291	16	34
Bloomberg	21,853	15	49
New York Times	21,092	14	63
USA Today	14,123	9	72

This table reports popular news sources of articles shared using Bitly. Our sample period is from January 2012 to July 2018, which includes a total of 79 nonfarm payroll announcements. In panel A, we consider clicks 2 hours prior to the release of the nonfarm payroll announcement, from 6:29 a.m. to 8:29 a.m. ET. There are a total of 21,283 clicks during this period across the 79 announcements. In panel B, we consider clicks during and after the announcement, from 8:30 a.m. to 10:30 a.m. ET. There are a total of 148,971 clicks during this period across the 79 announcements. The nonfarm payroll announcement is released by the Bureau of Labor Statistics at 8:30 a.m. ET on the first Friday of the month.

(a) Table 1: popular news sources

Table 2
Who shares Bitly links?

Bitly user type	Number of clicks	Percent of total number of clicks	Cumulative percent
<i>A. Prior to nonfarm payroll release, from 6:29 a.m. ET to 8:29 a.m. ET</i>			
Official WSJ Users	4,755	22	22
Official Bloomberg Users	4,609	22	44
Official CNN Users	3,683	17	61
Three Individual Users	1,672	8	69
Anonymous	1,029	5	74
Official USA Today Users	965	5	79
Official NY Times Users	781	4	82
<i>B. During and after nonfarm payroll release, from 8:30 a.m. ET to 10:30 a.m. ET</i>			
Official WSJ Users	18,747	13	13
Official NY Times Users	16,880	11	24
Official CNN Users	16,152	11	35
Official Bloomberg Users	15,295	10	45
Official USA Today Users	12,756	9	54
Three Individual Users	11,731	8	61
Anonymous	10,437	7	68

This table reports who shares Bitly links. Our sample period is from January 2012 to July 2018, which includes a total of 79 nonfarm payroll announcements. In panel A, we consider clicks 2 hours prior to the release of the nonfarm payroll announcement, from 6:29 a.m. to 8:29 a.m. ET. There are a total of 21,283 clicks during this period across the 79 announcements. In panel B, we consider clicks during and after the announcement, from 8:30 a.m. to 10:30 a.m. ET. There are a total of 148,971 clicks during this period across the 79 announcements. The nonfarm payroll announcement is released by the Bureau of Labor Statistics at 8:30 a.m. ET on the first Friday of the month. We aggregate clicks on links shared by three different individual users. News services often have more than one Bitly user account. In general, one Bitly user accounts for the majority of the clicks, but we aggregate across official users within a news service. The list of official usernames per news service was provided to us by Bitly.

(b) Table 2: who shares Bitly links?

5.3 Table 3 and Figures 3–4: relation to other uncertainty proxies

Read:

- Table 3 shows that Bitly demand is positively correlated with swap-based monetary-policy uncertainty, news-based uncertainty, VIX, forecast error, Treasury realized volatility, Google Trends, and information supply.
- The correlation with dispersion of professional forecasts is negative and insignificant.
- Figure 3 compares Bitly counts with Google Trends; the two series co-move but are not identical.
- Figure 4 compares Bitly counts with realized Treasury volatility; both rise around major uncertainty episodes such as Brexit and other macro events.

Economic message: Bitly demand is related to standard uncertainty proxies but is not redundant. It captures a demand-side dimension of attention to macro news.

Table 3
Contemporaneous relation between information demand and uncertainty measures

	Demand	Unc.S.	Unc.N.	VIX	Error	Disp.	Vol.	Google
Demand	1							
Unc.S.	0.38***	1						
Unc.N.	0.32**	0.55***	1					
VIX	0.22	0.44***	0.28*	1				
Error	0.25*	0.11	0.06	0.10	1			
Disp.	-0.15	-0.15	-0.18	-0.03	0.24*	1		
Vol.	0.36**	0.65***	0.52***	0.15	-0.0001	-0.34**	1	
Google	0.46***	0.35**	0.15	0.22*	0.27*	0.05	0.06	1
Supply	0.38***	-0.19	-0.15	0.31**	0.14	0.14	-0.46***	0.46***

This table reports the contemporaneous correlation between monthly information demand and monthly measures of uncertainty using data from January 2012 to July 2018. Demand is our measure of information demand (Bitly count). Unc.S. is a market-based measure of uncertainty that uses interest rate swaps. Unc.N. is a news-based measure of uncertainty. VIX is the VIX index. Error is the absolute value of forecast error. Disp. is the dispersion across professional forecasts. Vol. is the 2-year U.S. Treasury realized volatility. Google is the Google index for the nonfarm payroll topic; and Supply is the media news count for the topic nonfarm payroll using RavenPack. $***p < 0.01$, $**p < 0.05$, $*p < 0.1$.

(a) Table 3: correlations

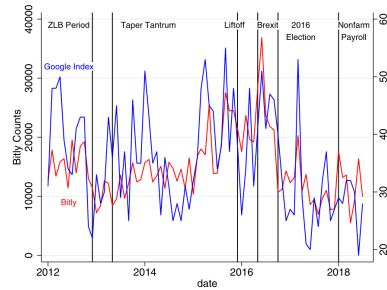


Figure 3
Bitly counts and the Google Trends index
The figure shows monthly Bitly counts (red line) and the Google Trends index (blue line) for the topic of nonfarm payroll announcements in our sample from January 2012 to July 2018.

(b) Figure 3: Bitly and Google

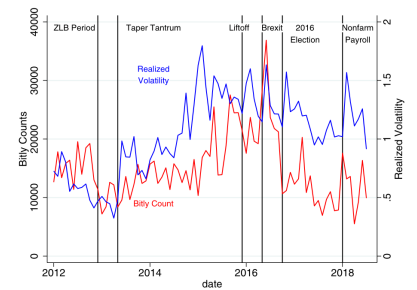


Figure 4
Bitly counts and 2-year U.S. Treasury volatility
The figure shows monthly Bitly counts (red line) and 2-year U.S. Treasury volatility (blue line).

(c) Figure 4: Bitly and realized volatility

5.4 Tables 4 and 5: baseline response and the main Bitly interaction

Read:

- Table 4 confirms that standardized nonfarm payroll surprises significantly move 2-, 5-, and 10-year Treasury futures yields.
- In the January 2012–July 2018 sample, the NFP surprise coefficients are 3.188 bps for 2-year notes, 6.437 bps for 5-year notes, and 7.215 bps for 10-year notes.
- Table 5 is the main result: NFP surprise $\times$ Bitly count equals 3.001, 3.521, and 2.728 for 2-, 5-, and 10-year notes.
- The high-Bitly dummy results are also positive: 4.213, 5.131, and 4.046.

Economic message: The same news surprise has a larger price impact when investors searched more before the announcement.

Table 4
U.S. Treasury futures response to nonfarm payroll surprises

	January 2004–July 2018	January 2012–July 2018	
	(1)	(2)	(3)
<i>A. Response of the 2-year U.S. Treasury note futures</i>			
Nonfarm payroll (NFP) surprise	4.953*** (0.606)	3.188*** (0.701)	3.584*** (0.819)
NFP surprise $\times$ SW ZLB period			−2.373** (0.920)
SW ZLB period			−1.076 (0.646)
Constant	0.632 (0.441)	0.0648 (0.452)	0.249 (0.526)
Number of obs.	175	79	79
Adjusted R^2	.335	.233	.259
<i>B. Response of the 5-year U.S. Treasury note futures</i>			
Nonfarm payroll (NFP) surprise	5.951*** (0.735)	6.437*** (0.992)	6.526*** (1.151)
NFP surprise $\times$ SW ZLB period			−0.248 (1.746)
SW ZLB period			−2.656** (1.285)
Constant	0.371 (0.514)	−0.144 (0.696)	0.261 (0.803)
Number of obs.	175	79	79
Adjusted R^2	.339	.343	.360
<i>C. Response of the 10-year U.S. Treasury note futures</i>			
Nonfarm payroll (NFP) surprise	5.881*** (0.727)	7.215*** (1.060)	6.790*** (1.180)
NFP surprise $\times$ SW ZLB period			3.038 (2.137)
SW ZLB period			−3.062* (1.617)
Constant	0.472 (0.506)	−0.180 (0.730)	0.259 (0.820)
Number of obs.	175	79	79
Adjusted R^2	.342	.374	.399

This table reports estimates of Equation (14) using two different samples. In column 1, the sample is from January

Table 5
Impact of information demand on the U.S. Treasury futures response to nonfarm payroll surprises

	(1)	(2)	(3)	(4)	(5)	(6)
	Two-year		Five-year		Ten-year	
NFP surprise (NFPs)	0.244 (0.674)	0.918 (0.607)	2.978** (1.223)	3.636*** (1.230)	4.514*** (1.437)	4.959*** (1.441)
NFPs $\times$ BC			3.001*** (0.834)	3.521*** (1.140)	2.728** (1.211)	
Bitly count (BC)			1.028* (0.593)	1.240 (0.798)	1.133 (0.769)	
NFPs $\times$ High BC			4.213*** (1.244)	5.131*** (1.880)		4.046* (2.075)
High Bitly count			0.393 (0.861)	0.705 (1.321)		0.836 (1.397)
Constant	−1.009** (0.446)	−0.459 (0.357)	−1.433* (0.758)	−0.896 (0.685)	−1.329 (0.845)	−0.914 (0.796)
Number of Obs.	79	79	79	79	79	79
Adjusted R^2	.448	.331	.451	.397	.435	.404

This table reports estimates of the response of U.S. Treasury futures on 2-, 5-, and 10-year notes to nonfarm payroll surprises using data from January 2012 to July 2018. The dependent variable is a 30-minute U.S. Treasury futures yield change using the prevailing futures price as of one second before the announcement to 30 minutes after the announcement. The variable Bitly count (BC) is the sum of clicks on newspaper articles related to nonfarm payroll from 2 hours before the release of the announcement to 1 minute prior to the announcement. We divide Bitly count by its standard deviation so that the magnitude of the coefficient can be interpreted more easily. The variable High Bitly count (High BC) is an indicator variable equal to one if the Bitly counts are above the median number of counts. Robust standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 4: baseline NFP response

(b) Table 5: Bitly demand and NFP response

5.5 Tables 6–8: horse races with other measures

Read:

- Table 6 shows that for 2-year notes, Bitly demand remains significant after controls: NFP surprise $\times$ Bitly count is 3.475 in the full control specification.
- Table 7 shows similar results for 5-year notes: the full specification coefficient is 4.270.
- Table 8 shows similar results for 10-year notes: the full specification coefficient is 3.820.
- Google search is not significant in the horse-race specifications, while information supply is sometimes significant but does not eliminate the Bitly effect.

Economic message: Bitly clicks contain unique information about uncertainty that is not fully captured by conventional uncertainty proxies or information supply.

Table 6
Response of the 2-year U.S. Treasury futures to nonfarm payroll surprises

	(1)	(2)	(3)	(4)	(5)	(6)
NFP surprise (NFPs)	-1.690 (2.623)	5.452* (2.861)	6.254*** (2.190)	1.003 (1.501)	-2.193 (2.070)	-0.380 (7.218)
	Monetary policy variables					
NFPS × FFTR	-3.808*** (0.888)					4.920 (4.174)
NFPS × ZLB	-1.898** (0.835)					1.958 (1.805)
NFPS × Unc. (swaps)	2.704*** (0.959)					5.701*** (2.455)
NFPS × Unc. (news)	-0.0131 (0.0157)					-0.00734 (0.0223)
	Risk					
NFPS × VIX		-0.150 (0.175)				-0.382** (0.144)
	Information environment					
NFPS × Noise			0.0195			0.0425 (0.0626)
NFPS × Error			0.00766			0.009644 (0.0243)
NFPS × Dispersion			-0.111*** (0.112)			-0.0894 (0.118)
	Trading volume and realized volatility					
NFPS × Volume						0.711 (0.600)
NFPS × Volatility						-0.811 (0.986)
	Information demand and supply					
NFPS × BC						2.599*** (0.830)
NFPS × Google						3.475*** (1.291)
NFPS × Supply						-0.772 (0.786)
Constant	-1.264 (2.097)	2.439 (2.015)	0.732 (1.335)	-2.151* (1.231)	-0.989 (1.554)	0.704 (4.837)
Number of obs.	79	79	79	79	79	79
Adjusted R ²	.420	.250	.309	.299	.490	.624

This table reports estimates of the response of U.S. Treasury futures on 2-year notes to nonfarm payroll surprises using data from January 2012 to July 2018. The dependent variable is a 30-minute U.S. Treasury futures yield change using the prevailing futures price as of one second before the announcement to 30 minutes after the announcement. All the interaction terms are measured before the release of the announcement. The estimation also includes main effects, but we do not report these coefficients. The variables Bitly count, Google index, and media coverage count are divided by their standard deviation so that the magnitude of the coefficient can be

(a) Table 6: 2-year controls

Table 7
Response of the 5-year U.S. Treasury futures to nonfarm payroll surprises

	(1)	(2)	(3)	(4)	(5)	(6)
NFP surprise (NFPs)	7.104 (5.081)	8.466* (4.516)	8.693*** (2.955)	0.0774 (4.835)	-2.599 (3.034)	5.546 (12.08)
	Monetary policy variables					
NFPS × FFTR						-2.782 (5.363)
NFPS × ZLB						1.320 (3.162)
NFPS × Unc. (swaps)						-1.176 (2.793)
NFPS × Unc. (news)						-0.0264 (0.0361)
	Risk					
NFPS × VIX		-0.134 (0.281)				-0.271 (0.275)
	Information environment					
NFPS × Noise			0.0709			0.0282 (0.0855)
NFPS × Error			-0.0166			-0.00873 (0.0416)
NFPS × Dispersion			-0.258 (0.161)			-0.114 (0.194)
	Trading volume and realized volatility					
NFPS × Volume						0.744** (0.327)
NFPS × Volatility						0.897 (2.266)
	Information demand and supply					
NFPS × BC						3.545*** (1.158)
NFPS × Google						4.270** (1.771)
NFPS × Supply						-1.800 (1.245)
Constant	-0.402 (3.661)	3.610 (3.076)	0.787 (1.988)	-3.535 (3.140)	-1.596 (2.508)	0.0885 (9.181)
Number of obs.	79	79	79	79	79	79
Adjusted R ²	.485	.356	.379	.410	.528	.619

This table reports estimates of the response of U.S. Treasury futures on 5-year notes to nonfarm payroll surprises using data from January 2012 to July 2018. The dependent variable is a 30-minute U.S. Treasury futures yield change using the prevailing futures price as of one second before the announcement to 30 minutes after the announcement. All the interaction terms are measured before the release of the announcement. The estimation also includes main effects, but we do not report these coefficients. The variables Bitly count, Google index, and media coverage count are divided by their standard deviation so that the magnitude of the coefficient can be

(b) Table 7: 5-year controls

Table 8
Response of the 10-year U.S. Treasury futures to nonfarm payroll surprises

	(1)	(2)	(3)	(4)	(5)	(6)
NFP surprise (NFPs)	11.11* (5.597)	7.040 (4.637)	6.693** (3.088)	0.786 (8.333)	-3.045 (3.277)	9.790 (14.97)
	Monetary policy variables					
NFPS × FFTR						-8.614*** (1.941)
NFPS × ZLB						1.722 (2.693)
NFPS × Unc. (swaps)						-1.219 (1.519)
NFPS × Unc. (news)						-0.0290 (0.0377)
	Risk					
NFPS × VIX		0.0130 (0.293)				-0.266 (0.296)
	Information environment					
NFPS × Noise			0.131*			0.0752 (0.0669)
NFPS × Error			-0.0361			-0.0181 (0.0399)
NFPS × Dispersion			-0.0717 (0.182)			-0.540 (0.243)
	Trading volume and realized volatility					
NFPS × Volume						0.537*** (0.248)
NFPS × Volatility						1.527 (1.660)
	Information demand and supply					
NFPS × BC						2.659** (1.279)
NFPS × Google						3.820** (1.604)
NFPS × Supply						-1.512 (1.585)
Constant	0.452 (4.280)	3.943 (3.191)	0.577 (2.057)	-3.537 (4.951)	-0.960 (2.738)	3.489 (10.97)
Number of obs.	79	79	79	79	79	79
Adjusted R ²	.508	.388	.406	.438	.523	.624

This table reports estimates of the response of U.S. Treasury futures on 10-year notes to nonfarm payroll surprises using data from January 2012 to July 2018. The dependent variable is a 30-minute U.S. Treasury futures yield change using the prevailing futures price as of one second before the announcement to 30 minutes after the announcement. All the interaction terms are measured before the release of the announcement. The estimation also includes main effects, but we do not report these coefficients. The variables Bitly count, Google index, and media coverage count are divided by their standard deviation so that the magnitude of the coefficient can be

(c) Table 8: 10-year controls

5.6 Tables 9 and 10: high-frequency timing and order flow

Read:

- Table 9 compares Bitly demand measured 2 hours before, 1 day before, and 1 month before the announcement.
- The 2-hour measure has the strongest predictive power in the joint specification, consistent with the value of high-frequency clickstream data.
- Table 10 adds order-flow information. The NFP surprise interaction with Bitly count remains positive, and order-flow effects are also stronger when high Bitly counts indicate high uncertainty.

Economic message: Information demand measured close to the announcement is more informative than lower-frequency proxies. Order flow does not absorb the Bitly result.

Table 9
Impact of information demand aggregated at different nonoverlapping frequencies on the U.S. Treasury futures response to nonfarm payroll

	(1)	(2)	(3)	(4)
<i>A. Two-year U.S. Treasury note futures</i>				
Nonfarm payroll surprise (NFPS)	0.244 (0.674)	0.099 (0.800)	-0.552 (0.671)	-0.632 (0.687)
NFPS × BC (2 hours before)	3.001*** (0.834)			2.252** (0.959)
NFPS × BC (1 day before)		2.104*** (0.637)		-0.0186 (0.780)
NFPS × BC (1 month before)			2.979*** (0.603)	1.215 (1.105)
Constant	-1.009** (0.446)	-0.808 (0.677)	-0.723 (0.439)	-0.741 (0.601)
Number of Obs.	79	79	79	79
Adjusted R^2	.448	.351	.392	.487
<i>B. Five-year U.S. Treasury note futures</i>				
Nonfarm payroll surprise	2.978** (1.223)	3.023** (1.426)	3.279** (1.511)	2.906* (1.532)
NFPS × BC (2 hours before)	3.521*** (1.140)			3.279** (1.365)
NFPS × BC (1 day before)		2.333** (0.960)		1.073 (1.352)
NFPS × BC (1 month before)			2.495** (1.012)	-1.144 (1.963)
Constant	-1.433* (0.758)	-0.991 (1.086)	-0.555 (0.820)	-0.405 (1.072)
Number of obs.	79	79	79	79
Adjusted R^2	.451	.395	.387	.473
<i>C. Ten-year U.S. Treasury note futures</i>				
Nonfarm payroll surprise	4.514*** (1.437)	4.288*** (1.619)	5.138*** (1.796)	4.673** (1.846)
NFPS × BC (2 hours before)	2.728** (1.211)			2.588* (1.479)
NFPS × BC (1 day before)		2.005* (1.050)		1.872 (1.542)
NFPS × BC (1 month before)			1.630 (1.069)	-2.331 (2.180)
Constant	-1.329 (0.845)	-0.826 (1.135)	-0.315 (0.952)	-0.083 (1.184)
Number of obs.	79	79	79	79
Adjusted R^2	.435	.407	.392	.461

This table reports the response of 2-, 5-, and 10-year U.S. Treasury note futures to nonfarm payroll surprises using data from January 2012 to July 2018. The variable BC (Bitly count 2 hours before) is the sum of clicks

(a) Table 9: frequency of information demand

Table 10
Order flow impact

	(1)	(2)	(3)	(4)	(5)	(6)
		Two-year	Five-year	Ten-year		
NFP surprise (NFPS)	1.279*** (0.067)	1.318*** (0.067)	3.999*** (0.101)	4.056*** (0.102)	3.912*** (0.106)	4.262*** (0.107)
NFPS × BC	1.988*** (0.050)	1.940*** (0.050)	1.843*** (0.076)	1.774*** (0.077)	1.237*** (0.077)	0.816*** (0.080)
Constant	0.0003 (0.002)	0.0003 (0.002)	-0.0018 (0.003)	-0.0017 (0.003)	-0.0023 (0.003)	-0.0026 (0.003)
Two hours before						
Order flow	0.636*** (0.054)	0.495*** (0.063)	1.257*** (0.098)	1.242*** (0.139)	1.210*** (0.064)	1.149*** (0.087)
Order flow × HBC		0.506*** (0.120)		0.0305 (0.196)		0.131 (0.128)
Two hours after						
Order flow	1.288*** (0.027)	1.041*** (0.038)	2.381*** (0.042)	2.154*** (0.060)	1.867*** (0.023)	1.499*** (0.030)
Order flow × HBC		0.517*** (0.054)		0.432*** (0.083)		0.797*** (0.044)
Number of Obs.	18,960	18,960	18,960	18,960	18,960	18,960
Adjusted R^2	.327	.331	.404	.405	.484	.493

This table reports the response of U.S. Treasury futures to nonfarm payroll surprises and order flow using data from January 2012 to July 2018. The dependent variable is the 1-minute U.S. Treasury futures yield change using the prevailing futures yield as of the end of the minute. Order flow is estimated using the Lee and Ready

(b) Table 10: order-flow impact

5.7 Figure 5 and Table 11: no reversal or delayed adjustment

Read:

- Figure 5 shows that yield changes at the announcement are much larger on high-click days.
- The figure does not show obvious post-announcement reversal or drift.
- Table 11 formalizes this point with lead-lag regressions.
- The contemporaneous surprise and surprise × Bitly terms are significant, while the lag and lead sums are not.

Economic message: High information demand does not simply indicate noise attention or predictable reversal. It is more consistent with larger uncertainty being resolved at the announcement.

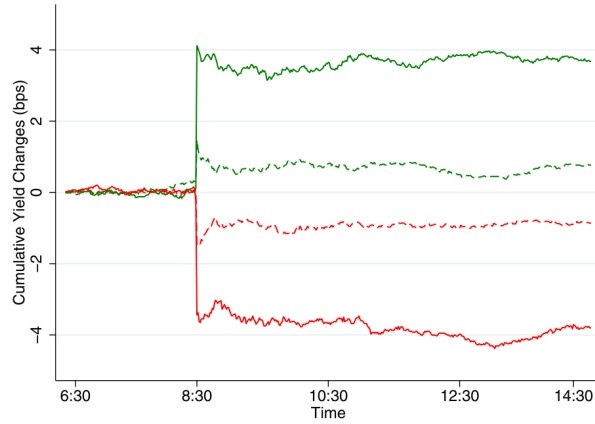


Figure 5
intraday 2-year U.S. Treasury yield reaction
The figure shows the aggregate intraday reaction of the 2-year U.S. Treasury futures yields to nonfarm payroll surprises across 79 announcement days from January 2012 to July 2018. We perform a dependent sort. First, we sort on positive (green lines) and negative (red lines) nonfarm payroll surprises, and then we sort on Bitly counts in the top 50% percentile (solid lines) and the bottom 50% percentile (dashed lines). The nonfarm payroll

(a) Figure 5: intraday 2-year yield reaction

Table 11
Pre- and post-announcement reaction

	(1)	(2)	(3)	(4)	(5)	(6)
	Two-year		Five-year		Ten-year	
NFPS(t-1)	-0.179 (0.498)	-0.182 (0.689)	-0.876 (1.944)	-1.218 (2.721)	-1.366 (3.344)	-1.556 (4.695)
Sum of 30 lagged NFPS coeff.	-4.035	-1.153	-16.527	-7.672	-26.149	-15.605
F-statistic	1.926	0.010	2.045	0.131	1.644	0.260
NFPS(t-1)×BC(t-1)		0.0224 (0.516)		0.311 (2.039)		0.242 (3.518)
Sum of 30 lagged NFPS × BC coeff.		-2.915		-9.031		-10.675
F-statistic						
NFPS(t)	3.692*** (0.498)	1.308* (0.690)	15.85*** (1.944)	10.08*** (2.725)	28.64*** (3.345)	22.52*** (4.702)
NFPS(t)×BC		2.449*** (0.520)		6.156*** (2.053)		6.940* (3.541)
NFPS(t+1)	-0.419 (0.498)	-0.540 (0.689)	-1.243 (1.944)	-1.241 (2.721)	-1.563 (3.344)	-1.548 (4.694)
Sum of 30 lead NFPS coeff.	-0.825	-1.562	0.741	-7.587	9.147	-3.768
F-statistic	0.107	0.216	0.001	0.381	0.218	0.085
NFPS(t+1)×BC		0.126 (0.516)		-0.00198 (2.038)		0.0262 (3.516)
Sum of 30 lead NFPS × BC coeff.		0.664		7.997		12.354
F-statistic						
Constant	0.0929 (0.0623)	0.091 (0.0621)	0.102 (0.243)	0.740 (0.245)	-0.0214 (0.418)	0.685 (0.423)
Number of obs.	2,357	2,357	2,357	2,357	2,357	2,357
Adjusted R ²	.040	.090	.043	.071	.047	.070

This table reports the response of U.S. Treasury futures prices to nonfarm payroll announcements using data from January 2012 to July 2018. The dependent variable is 1-day U.S. Treasury futures yield changes using the prevailing futures yield as of 4:00 p.m. ET. Standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Next, to analyze this point more formally, we estimate the following equation

(b) Table 11: pre/post reaction

5.8 Tables 12 and 13: other macro announcements and FOMC news

Read:

- Table 12 shows that the Bitly interaction is positive for retail sales and ISM manufacturing PMI announcements.
- Table 13 shows that FOMC path surprises have a larger impact when information demand is high.
- The target surprise interaction is not significant, which fits the idea that the relevant uncertainty is about the future path of rates rather than only the current target change.

Economic message: The evidence generalizes beyond nonfarm payroll. The strongest FOMC result is for path surprises, exactly the component most related to future interest rates.

Table 12
Other important macroeconomic news announcements

	(1)	(2)	(3)	(4)
	Retail sales		ISM manufacturing PMI	
Surprise	1.094*** (0.223)	0.725** (0.287)	0.427** (0.210)	0.371* (0.188)
Surprise x Bitly count		0.581** (0.284)		0.369*** (0.132)
Bitly count		-0.194 (0.237)		0.0720 (0.129)
Constant	-0.199 (0.231)	-0.102 (0.287)	0.206 (0.187)	0.104 (0.178)
Number of obs.	79	79	79	79
Adjusted R ²	.238	.278	.301	.334

This table reports the response of U.S. Treasury futures on 2-year notes to ISM and Retail Sales surprises using data from January 2012 to July 2018. The dependent variable is a 30-minute U.S. Treasury futures yield change using the prevailing futures price as of one second before the announcement to 30 minutes after the announcement. The variables Bitly count is the sum of clicks on newspaper articles related to ISM or Retail Sales from 2 hours before the release of the announcement to 1 minute prior to the announcement. We divide Bitly Count by its standard deviation so that the magnitude of the coefficient can be interpreted more easily. Robust standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$

(a) Table 12: other macro announcements

Table 13
Two-year yield response to FOMC surprises

	(1)	(2)	(3)	(4)	(5)	(6)
Target surprise (TS)	1.281*** (0.233)	1.955** (0.858)				0.774*** (0.226)
TS×Bitly count		-0.278 (0.298)				-0.0370 (0.169)
TS×Press conference						0.0515 (0.563)
Path surprise (PS)			1.136*** (0.0486)	0.930*** (0.0953)	0.939*** (0.0999)	1.047*** (0.0783)
PS×Bitly count				0.268** (0.126)	0.261* (0.131)	0.154* (0.0819)
PS×Press conference					0.0127 (0.0749)	-0.0778 (0.0749)
Bitly count		0.448 (0.477)		-0.0459 (0.361)	-0.114 (0.384)	0.113 (0.130)
Press conference					0.418 (0.356)	0.308 (0.301)
Constant	-0.140 (0.562)	-0.348 (0.582)	-0.431** (0.199)	-0.420* (0.237)	-0.581*** (0.209)	-0.460*** (0.164)
Number of obs.	52	52	52	52	52	52
Adjusted R ²	.140	.159	.893	.913	.915	.954

This table reports the response of U.S. Treasury futures on 2-year notes to monetary policy surprises using data from January 2012 to July 2018. The dependent variable is a 2-hour U.S. Treasury futures yield change using the prevailing futures price as of 15 minutes before the announcement to 1 hour and 45 minutes after the

(b) Table 13: FOMC path and target surprises

6 Short summary

- The paper proposes information demand as a proxy for investor uncertainty.
- The model links uncertainty, information acquisition, and price sensitivity to public news.
- Bitly clickstream data provide a high-frequency measure of information demand before macro announcements.
- The main sample studies nonfarm payroll announcements from January 2012 to July 2018.
- Clicks rise before the 8:30 a.m. release and spike after the release.
- The clicked articles are mostly from major financial news providers, especially WSJ and Bloomberg.
- Bitly demand is correlated with standard uncertainty proxies but contains additional information.
- Higher pre-announcement Bitly demand predicts a larger response of 2-, 5-, and 10-year Treasury futures yields to NFP surprises.
- The result survives horse races with market uncertainty, VIX, realized volatility, Google Trends, information supply, and order flow.
- The predictive power is strongest for high-frequency clicks measured close to the announcement.
- High information demand does not predict reversal or underreaction after the announcement.
- Similar patterns appear for retail sales, ISM manufacturing PMI, and FOMC path surprises.

7 Conclusion

7.1 Core conclusion

Benamar, Foucault, and Vega show that investors' information demand can be used as a proxy for uncertainty. In Treasury markets, more pre-announcement demand for nonfarm payroll information predicts a larger yield reaction to the payroll surprise.

7.2 Economic intuition

When investors are uncertain about the future path of interest rates, the marginal value of information rises. They click on more relevant news before announcements, and once the announcement arrives, it resolves more uncertainty. Therefore, Treasury prices move more strongly in response to the same standardized surprise.

7.3 Takeaway

The paper's broader message is methodological: clickstream data can measure uncertainty from the demand side. Standard uncertainty proxies often measure volatility, disagreement, or news supply. Bitly clicks reveal what investors choose to learn about before payoff-relevant news arrives.